

BCSL-013: COMPUTER BASICS AND PC SOFTWARE LAB

Assignment Solutions (2025-26)

Q1: Linux Operating System Commands (16 Marks)

Part (a): Linux Commands with Screenshots (10 Marks)

Instructions: Execute each command in Linux terminal and capture screenshots.

i) date command

Purpose: Displays or sets the system date and time

Command to execute:

```
date
```

Sample Output:

```
Thu Oct  5 14:32:45 IST 2025
```

Additional variations to try:

```
date +%d/%m/%Y          # Display date in DD/MM/YYYY format
date +%A                 # Display day of week
date +%T                 # Display time in HH:MM:SS format
date "+%Y-%m-%d %H:%M:%S" # Custom format
```

Screenshot should show: The current date and time in the terminal

ii) man command

Purpose: Displays the manual page for any command

Command to execute:

```
man ls
```

Sample Output: Opens the manual page for 'ls' command showing:

- NAME section
- SYNOPSIS section
- DESCRIPTION section
- OPTIONS section

Additional variations:

```
man date           # Manual for date command
man -k copy        # Search for commands related to 'copy'
man 5 passwd       # Manual for passwd file (section 5)
man -f ls          # Short description of ls
```

How to navigate man pages:

- Press Space - Next page
- Press b - Previous page
- Press /word - Search for 'word'
- Press q - Quit

Screenshot should show: The man page opened in terminal

iii) mv command

Purpose: Moves or renames files and directories

Commands to execute:

```
# Create a test file
echo "This is a test file" > testfile.txt

# Rename a file
mv testfile.txt newfile.txt

# Verify the rename
ls -l newfile.txt

# Move file to another directory (assuming Documents exists)
mkdir -p ~/test_directory
mv newfile.txt ~/test_directory/

# Verify the move
ls -l ~/test_directory/
```

Additional variations:

```
mv file1.txt file2.txt file3.txt /destination/ # Move multiple files
mv -i oldname.txt newname.txt                 # Interactive (asks
before overwrite)                             # before overwrite)
mv -v oldname.txt newname.txt                  # Verbose (shows what's
being done)                                    # being done)
mv *.txt /destination/                        # Move all .txt files
```

Screenshot should show: Commands and their outputs showing file being moved/renamed

iv) ln command

Purpose: Creates links (shortcuts) between files

Commands to execute:

```
# Create a file
echo "Original file content" > original.txt

# Create a hard link
ln original.txt hardlink.txt

# Create a symbolic (soft) link
ln -s original.txt softlink.txt

# List files with inode numbers
ls -li original.txt hardlink.txt softlink.txt

# Display content of links
cat hardlink.txt
cat softlink.txt
```

Explanation:

- **Hard link:** Same inode number, same file content, changes reflect in both
- **Soft link:** Different inode, acts as pointer, shown with -> symbol

Additional variations:

```
ln -s /path/to/file linkname      # Create symbolic link
ln -f file1 file2                  # Force creation, overwrite if exists
ln -v file1 file2                  # Verbose output
```

Screenshot should show: ls -li output showing inode numbers and link types

v) chmod command

Purpose: Changes file permissions (read, write, execute)

Commands to execute:

```
# Create a test file
echo "Permission test file" > permtest.txt

# Check current permissions
ls -l permtest.txt

# Change permissions using symbolic notation
chmod u+x permtest.txt      # Add execute for user
```

```
ls -l permtest.txt

chmod go-w permtest.txt          # Remove write for group and others
ls -l permtest.txt

# Change permissions using numeric notation
chmod 755 permtest.txt           # rwxr-xr-x
ls -l permtest.txt

chmod 644 permtest.txt           # rw-r--r--
ls -l permtest.txt

chmod 777 permtest.txt           # rwxrwxrwx
ls -l permtest.txt
```

Permission codes:

- **r (read) = 4**
- **w (write) = 2**
- **x (execute) = 1**

Common permissions:

- 755 = rwxr-xr-x (owner: all, group/others: read+execute)
- 644 = rw-r--r-- (owner: read+write, group/others: read only)
- 777 = rwxrwxrwx (all permissions for everyone)

Screenshot should show: Permission changes using ls -l

vi) find command

Purpose: Searches for files and directories in a directory hierarchy

Commands to execute:

```
# Find files by name in current directory
find . -name "*.txt"

# Find files by name (case-insensitive)
find . -iname "test*"

# Find files modified in last 7 days
find . -mtime -7

# Find files larger than 1MB
find . -size +1M

# Find files and execute command on them
find . -name "*.txt" -exec ls -lh {} \;

# Find empty files
find . -type f -empty

# Find directories only
```

```
find . -type d

# Find with multiple conditions
find . -name "*.txt" -size +1k -mtime -30
```

Additional examples:

```
find /home -name "document.pdf"          # Search in specific directory
find . -user username                    # Find files owned by user
find . -perm 644                          # Find files with specific
permissions
```

Screenshot should show: Find command output listing matching files

vii) diff command

Purpose: Compares two files line by line

Commands to execute:

```
# Create two files with slight differences
echo -e "Line 1\nLine 2\nLine 3\nLine 4" > file1.txt
echo -e "Line 1\nLine 2 modified\nLine 3\nLine 5" > file2.txt

# Compare files
diff file1.txt file2.txt

# Side-by-side comparison
diff -y file1.txt file2.txt

# Unified format (like git diff)
diff -u file1.txt file2.txt

# Ignore case differences
diff -i file1.txt file2.txt

# Brief output (just whether files differ)
diff -q file1.txt file2.txt
```

Understanding diff output:

- 2c2 means line 2 changed to line 2
- < indicates lines from first file
- > indicates lines from second file
- a means added, d means deleted, c means changed

Screenshot should show: Diff command output showing differences

viii) tail command

Purpose: Displays the last part of files

Commands to execute:

```
# Create a file with numbered lines
seq 1 50 > numbers.txt

# Display last 10 lines (default)
tail numbers.txt

# Display last 5 lines
tail -n 5 numbers.txt

# Display last 20 lines
tail -n 20 numbers.txt

# Display from line 40 onwards
tail -n +40 numbers.txt

# Follow file updates in real-time (useful for log files)
# Press Ctrl+C to stop
echo "Log entry 1" > logfile.log
tail -f logfile.log &
# In another terminal: echo "Log entry 2" >> logfile.log
```

Additional variations:

```
tail -c 50 file.txt                # Last 50 bytes
tail -n 5 file1.txt file2.txt      # Last 5 lines of multiple files
tail -f /var/log/syslog            # Follow system log (needs sudo)
```

Screenshot should show: Last few lines of a file

ix) pwd command

Purpose: Print Working Directory - shows current directory path

Commands to execute:

```
# Display current directory
pwd

# Change to home directory and show path
cd ~
pwd

# Change to root directory and show path
cd /
pwd

# Go back to previous directory
cd -
pwd

# Create nested directories and navigate
```

```
mkdir -p ~/test/level1/level2
cd ~/test/level1/level2
pwd
```

Additional information:

- `pwd` shows absolute path
- `pwd -P` shows physical path (resolves symbolic links)
- `pwd -L` shows logical path (may include symbolic links)

Screenshot should show: Current directory path

x) grep command

Purpose: Searches for patterns in files

Commands to execute:

```
# Create a sample file
cat > sample.txt << EOF
apple orange banana
Apple ORANGE Banana
The apple is red
oranges are orange
banana split dessert
EOF

# Search for 'apple' in file
grep "apple" sample.txt

# Case-insensitive search
grep -i "apple" sample.txt

# Search and show line numbers
grep -n "orange" sample.txt

# Search for whole word only
grep -w "orange" sample.txt

# Invert match (lines NOT containing pattern)
grep -v "apple" sample.txt

# Count matching lines
grep -c "orange" sample.txt

# Recursive search in directory
grep -r "pattern" /path/to/directory

# Search multiple files
grep "apple" *.txt
```

Additional variations:

```
grep -A 2 "pattern" file.txt          # Show 2 lines after match
```

```
grep -B 2 "pattern" file.txt      # Show 2 lines before match
grep -C 2 "pattern" file.txt      # Show 2 lines context
grep "^apple" file.txt            # Lines starting with 'apple'
grep "apple$" file.txt            # Lines ending with 'apple'
grep -E "apple|orange" file.txt   # Extended regex (OR condition)
```

Screenshot should show: Grep command searching and highlighting matches

Part (b): File Operations in Linux (6 Marks)

Step-by-step Solution:

Step 1: Create a text file with 10 lines

Command:

```
cat > myfile.txt << EOF
This is line one with a word
Line two contains a letter
A third line appears here
The fourth line has a character
Line five also has a word
Sixth line with a vowel
Line seven contains a symbol
A eighth line is here
The ninth line has a letter
This is the tenth and final line with a word
EOF
```

Alternative method using echo:

```
echo "This is line one with a word" > myfile.txt
echo "Line two contains a letter" >> myfile.txt
echo "A third line appears here" >> myfile.txt
echo "The fourth line has a character" >> myfile.txt
echo "Line five also has a word" >> myfile.txt
echo "Sixth line with a vowel" >> myfile.txt
echo "Line seven contains a symbol" >> myfile.txt
echo "A eighth line is here" >> myfile.txt
echo "The ninth line has a letter" >> myfile.txt
echo "This is the tenth and final line with a word" >> myfile.txt
```

Verify file creation:

```
cat myfile.txt
wc -l myfile.txt      # Should show 10 lines
```

Step 2: Create a copy of this file

Command:

```
cp myfile.txt myfile_copy.txt
```


Verify copy:

```
ls -l myfile.txt myfile_copy.txt
diff myfile.txt myfile_copy.txt    # Should show no difference
```

Step 3: Count how many times the word "a" appeared

Commands:

Method 1: Case-sensitive count (lowercase 'a' only)

```
grep -o "a" myfile.txt | wc -l
```

Method 2: Case-insensitive count (both 'a' and 'A')

```
grep -io "a" myfile.txt | wc -l
```

Method 3: Count whole word "a" only

```
grep -wo "a" myfile.txt | wc -l
```

Method 4: Using awk

```
awk '{for(i=1;i<=NF;i++) if($i=="a") count++} END {print count}' myfile.txt
```

Explanation:

- `grep -o` : prints only matching part
- `wc -l` : counts lines (number of matches)
- `-i` : case-insensitive
- `-w` : match whole word only

Expected output example:

```
Case-sensitive 'a': 8
Case-insensitive 'a' and 'A': 12
Whole word "a": 3
```

Step 4: Display file permissions

Command:

```
ls -l myfile.txt
```

Sample output:

```
-rw-rw-r-- 1 user user 245 Oct 5 15:30 myfile.txt
```

Understanding permissions:

```
-rw-rw-r--
| | | | |
| | | | |  Others: read only
| | | | |  Others: no write
| | | | |  Others: no execute
| | | | |  Group: read
| | | | |  Group: write
| | | | |  Group: no execute
| | | | |  Owner: read
| | | | |  Owner: write
| | | | |  Owner: no execute
| | | | |  File type (- = regular file)
```

Detailed display:

```
stat myfile.txt
```

Step 5: Change file permissions to rw-rw-rw-

Command using numeric notation:

```
chmod 666 myfile.txt
```

Command using symbolic notation:

```
chmod u=rw,g=rw,o=rw myfile.txt
```

Or step by step:

```
chmod u+rw myfile.txt      # User: read+write
chmod g+rw myfile.txt      # Group: read+write
chmod o+rw myfile.txt      # Others: read+write
chmod ugo-x myfile.txt     # Remove execute from all
```

Verify the change:

```
ls -l myfile.txt
```

Expected output:

```
-rw-rw-rw- 1 user user 245 Oct 5 15:30 myfile.txt
```

Complete Script for Part (b):

```
#!/bin/bash

echo "=== Step 1: Creating file with 10 lines ==="
cat > myfile.txt << EOF
This is line one with a word
Line two contains a letter
A third line appears here
The fourth line has a character
Line five also has a word
Sixth line with a vowel
```

Line seven contains a symbol
A eighth line is here
The ninth line has a letter
This is the tenth and final line with a word
EOF

```
echo -e "\n=== Verifying file creation ==="  
wc -l myfile.txt  
cat myfile.txt
```

```
echo -e "\n=== Step 2: Creating copy of file ==="  
cp myfile.txt myfile_copy.txt  
ls -l myfile*.txt
```

```
echo -e "\n=== Step 3: Counting occurrences of 'a' ==="  
echo "Case-sensitive 'a' count:"  
grep -o "a" myfile.txt | wc -l  
echo "Case-insensitive 'a' and 'A' count:"  
grep -io "a" myfile.txt | wc -l
```

```
echo -e "\n=== Step 4: Displaying file permissions ==="  
ls -l myfile.txt
```

```
echo -e "\n=== Step 5: Changing permissions to rw-rw-rw- ==="  
chmod 666 myfile.txt  
ls -l myfile.txt
```

```
echo -e "\n=== All tasks completed successfully ==="
```

Screenshots to attach:

1. File creation and content display
2. File copy creation and verification
3. Word count results
4. Original file permissions
5. Modified file permissions (rw-rw-rw-)

Q2: Presentation on "Importance of JavaScript in Web Development" (16 Marks)

Software Options:

- Microsoft PowerPoint
- LibreOffice Impress
- Google Slides
- WPS Presentation

I'll provide instructions for Microsoft PowerPoint:

Presentation Outline (Minimum 10 Slides)

Slide 1: Title Slide

- Title: "Importance of JavaScript in Web Development"
- Subtitle: "The Language that Powers the Modern Web"
- Your Name
- Date
- Image: JavaScript logo

Slide 2: What is JavaScript?

- Definition and brief history
- Created by Brendan Eich in 1995
- Client-side scripting language
- Image: Timeline of JavaScript evolution

Slide 3: Core Features of JavaScript

- Lightweight and interpreted
- Object-oriented programming
- Event-driven
- Platform-independent
- Image: JavaScript features diagram

Slide 4: JavaScript in Frontend Development

- DOM manipulation
- Dynamic content updates
- User interaction handling
- Form validation
- Image: Interactive webpage example

Slide 5: Popular JavaScript Frameworks

- React.js (Meta)
- Angular (Google)
- Vue.js
- Next.js
- Image: Framework logos

Slide 6: JavaScript in Backend Development

- Node.js platform
- Express.js framework
- Full-stack JavaScript development
- Image: Node.js architecture diagram

Slide 7: Importance in Modern Web Applications

- Single Page Applications (SPA)
- Progressive Web Apps (PWA)

- Real-time applications
- Image: Modern web app examples

Slide 8: JavaScript and APIs

- RESTful API consumption
- AJAX and Fetch API
- JSON data handling
- Image: API integration diagram

Slide 9: Career Opportunities

- High demand for JavaScript developers
- Salary ranges
- Job roles (Frontend, Backend, Full-stack)
- Image: Career growth chart

Slide 10: Future of JavaScript

- ECMAScript updates
- WebAssembly integration
- AI and ML integration
- Image: Future technology concepts

Step-by-Step Instructions to Create Presentation

Feature (i): Apply Common Design Template

Steps:

1. Open Microsoft PowerPoint
2. Click on **File** → **New**
3. In the search box, type "Professional" or "Technology"
4. Select a template (e.g., "Ion" or "Facet" theme)
5. Click **Create**

OR Apply theme to existing presentation:

1. Go to **Design** tab
2. Browse through **Themes** gallery
3. Click on desired theme (e.g., "Wisp", "Integral", "Ion")
4. Theme applies to all slides automatically

Alternative - Custom Theme:

1. Go to **Design** tab
2. Click on dropdown arrow in Themes section
3. Select **Browse for Themes**

4. Choose a .thmx file or create custom

Verification: All slides should have same background, fonts, and color scheme

Feature (ii): Add Audio and Video Files

Adding Audio File:

Steps:

1. Select a slide (e.g., Slide 5 about Frameworks)
2. Go to **Insert** tab
3. Click **Audio** → **Audio on My PC**
4. Browse and select an audio file (.mp3, .wav, .m4a)
5. Click **Insert**

Configure Audio Settings:

1. Click on the audio icon on slide
2. Go to **Playback** tab (Audio Tools)
3. Set **Start**:
 - **Automatically** (plays when slide appears)
 - **On Click** (plays when clicked)
4. Check **Hide During Show** (optional - hides icon during presentation)
5. Check **Loop until Stopped** (if desired)
6. Set **Volume** level

Alternative - Record Audio:

1. **Insert** → **Audio** → **Record Audio**
 2. Click **Record** button
 3. Speak into microphone
 4. Click **Stop** when done
 5. Name the recording and click **OK**
-

Adding Video File (Full-Screen Mode):

Steps:

1. Select a slide (e.g., Slide 7 about Modern Web Apps)
2. Go to **Insert** tab
3. Click **Video** → **Video on My PC**
4. Browse and select video file (.mp4, .avi, .mov)
5. Click **Insert**

Configure Video for Full-Screen:

1. Click on the video on slide
2. Go to **Playback** tab (Video Tools)
3. Check **Play Full Screen** option
4. Set **Start: Automatically** or **On Click**
5. Optional: Check **Loop until Stopped**
6. Optional: Check **Rewind after Playing**

Alternative - Trim Video:

1. Select video
2. **Playback** tab → **Trim Video**
3. Set start and end points
4. Click **OK**

Alternative - Online Video:

1. **Insert** → **Video** → **Online Video**
2. Paste YouTube URL or search
3. Click **Insert**

Note: For full-screen effect, you can also:

- Resize video to cover entire slide
 - Go to **Format** tab → **Video Styles** → **No Style**
 - Remove border for seamless look
-

Feature (iii): Timer-Based Transitions for All Slides

Steps to Add Transitions:

1. **Select First Slide** (or press Ctrl+A to select all slides from slide sorter view)
2. Go to **Transitions** tab
3. **Choose Transition Effect:**
 - Click on desired effect (e.g., "Push", "Wipe", "Fade", "Reveal")
 - Preview plays automatically
4. **Set Transition Duration:**
 - In **Timing** section
 - Set **Duration:** 01.00 seconds (or as desired)
5. **Enable Timer-Based Transition:**
 - In **Timing** section
 - **Uncheck** "On Mouse Click" (if you want automatic only)
 - **Check** "After" box
 - Set time: e.g., **00:05.00** (5 seconds)
 - This means slide will automatically advance after 5 seconds
6. **Apply to All Slides:**
 - Click **Apply To All** button
 - All slides now have same transition

Individual Slide Timing (if different timing needed):

1. Select specific slide
2. Set different "After" timing
3. Don't click "Apply To All"

Recommended Timing:

- Title slide: 3-5 seconds
- Content slides: 30-45 seconds
- Video slide: Duration of video + 2 seconds
- Conclusion slide: 5-10 seconds

Verify Transitions:

1. Go to **Slide Show** tab
 2. Click **From Beginning**
 3. Watch slides advance automatically
-

Feature (iv): Add Headings and Slide Notes

Adding Proper Headings:

For each slide:

1. Click on title placeholder
2. Type clear, descriptive heading
3. **Format heading:**
 - **Home** tab → **Font** section
 - Font: Arial or Calibri
 - Size: 40-44 pt for title
 - Bold: Yes
 - Color: Match theme or use contrasting color

Heading Examples:

- Slide 1: "Importance of JavaScript in Web Development"
 - Slide 2: "What is JavaScript?"
 - Slide 3: "Core Features of JavaScript"
 - Slide 4: "JavaScript in Frontend Development"
 - (Continue for all slides)
-

Adding Slide Notes:

Steps:

1. Switch to **Normal View** (if not already)
2. Look at bottom of screen - you'll see **Notes** section below slide
3. If Notes pane not visible:
 - Go to **View** tab
 - Check **Notes** in **Show** section
 - OR click **Notes** button at bottom
4. **Click in Notes pane** for each slide
5. **Type detailed notes** for each slide:

Example Notes:

Slide 1 Notes:

Introduction slide - Welcome audience and introduce the topic.
JavaScript is the most widely used programming language for web development.
It runs on 98% of all websites worldwide.
Key point: JavaScript makes websites interactive and dynamic.

Slide 2 Notes:

Explain JavaScript history:
- Created in just 10 days by Brendan Eich at Netscape
- Originally called "Mocha", then "LiveScript"
- Renamed to JavaScript for marketing purposes
- Not related to Java programming language
Key point: JavaScript has evolved significantly since 1995

Slide 3 Notes:

Discuss core features:
- Lightweight: Small memory footprint
- Interpreted: No compilation needed
- Object-oriented: Supports objects and inheritance
- Event-driven: Responds to user actions
- Platform-independent: Runs on any browser
Ask audience if they have experience with any programming language

(Continue detailed notes for remaining slides)

Slide 10 Notes:

Conclusion slide - Summarize key points:
1. JavaScript is essential for modern web development
2. Used for both frontend and backend
3. Huge ecosystem of frameworks and libraries
4. Excellent career opportunities
5. Continuously evolving with new features
Thank audience and open for questions

Formatting Notes:

- Keep notes concise but informative
- Include speaking points

- Add reminders for demonstrations
 - Note questions to ask audience
 - Include timing estimates
-

Feature (v): Add One Image per Slide

Steps to Insert Images:

Method 1: Insert from Computer

1. Select slide
2. Go to **Insert** tab
3. Click **Pictures** → **This Device**
4. Browse and select image
5. Click **Insert**

Method 2: Insert Online Pictures

1. **Insert** tab → **Pictures** → **Online Pictures**
2. Search for image (e.g., "JavaScript logo")
3. Select image
4. Click **Insert**

Method 3: Icons (Built-in)

1. **Insert** tab → **Icons**
2. Search for relevant icons
3. Select and insert

Resize and Position Images:

1. Click on image
2. Drag corner handles to resize (hold Shift for proportional)
3. Drag image to desired position
4. **Format** tab:
 - Apply **Picture Styles**
 - Add **Picture Border**
 - Add **Picture Effects** (shadow, reflection)

Image Recommendations for Each Slide:

- **Slide 1:** JavaScript logo (official JS logo in yellow and black)
- **Slide 2:** Brendan Eich photo or JavaScript history timeline
- **Slide 3:** Diagram showing JavaScript features in a circular layout
- **Slide 4:** Screenshot of interactive website or DOM tree diagram
- **Slide 5:** Logos of React, Angular, Vue.js arranged horizontally
- **Slide 6:** Node.js logo with server diagram
- **Slide 7:** Screenshots of popular web applications (Gmail, Facebook, Netflix)

- **Slide 8:** API integration diagram showing frontend-backend communication
- **Slide 9:** Career growth chart or developer at workstation
- **Slide 10:** Futuristic technology concept image

Image Sources (Free):

- Unsplash.com
- Pexels.com
- Pixabay.com
- Official framework websites for logos

Complete Sequence of Steps Summary

Phase 1: Initial Setup

1. Open PowerPoint → New Presentation
2. Select professional template
3. Save as "JavaScript_Importance.pptx"

Phase 2: Content Creation 4. Create 10 slides with titles and content 5. Add headings to each slide (formatted properly) 6. Write detailed notes for each slide

Phase 3: Multimedia Integration 7. Insert one relevant image per slide 8. Resize and position images aesthetically 9. Add audio file to Slide 5 (configured to auto-play) 10. Add video file to Slide 7 (configured for full-screen)

Phase 4: Transitions and Timing 11. Select all slides (Ctrl+A in Slide Sorter) 12. Go to Transitions tab 13. Choose transition effect (e.g., "Fade") 14. Set duration to 1 second 15. Enable "After" with 30-45 seconds timing 16. Click "Apply To All"

Phase 5: Review and Export 17. Run slideshow (F5) to test 18. Check audio and video playback 19. Verify transitions work smoothly 20. Make adjustments as needed

Phase 6: Export Handouts 21. Go to **File** → **Export** 22. Choose **Create Handouts** 23. Select **6 slides per page** 24. Export as PDF

Alternative Handout Method:

1. **File** → **Print**
2. Under **Settings**, select **Handouts**
3. Choose **6 Slides Horizontal**
4. Click **Print** or **Save as PDF**

Expected Deliverables:

1. **PowerPoint file** (.pptx) with all features implemented

2. **PDF handout** showing all 10 slides (6 per page, should be 2 pages)
 3. **Word document** with sequence of steps (this content)
 4. **Screenshots** showing:
 - Design template applied
 - Audio insertion
 - Video in full-screen settings
 - Transition timing settings
 - Slide notes example
 - Images inserted
-

Q3: Word Processing Tasks (16 Marks)

Part (a): Document about BCA Programme (6 Marks)

Required Components:

1. Multi-level list (courses and topics)
 2. Table (course details)
 3. Paragraph (objectives)
-

Step-by-Step Instructions

Step 1: Create New Document

1. Open Microsoft Word (or LibreOffice Writer)
 2. **File** → **New** → **Blank Document**
 3. Save as "BCA_Programme_Document.docx"
-

Step 2: Add Document Title

1. Type: "Bachelor of Computer Applications (BCA) Programme"
 2. **Select the title text**
 3. **Home** tab → **Styles** → Apply "Title" style
 4. OR manually format:
 - Font: Arial Black or Calibri Bold
 - Size: 20-22 pt
 - Alignment: Center
 - Color: Dark Blue
 5. Press Enter twice
-

Step 3: Create Multi-Level List

Section Heading:

1. Type: "Course Structure and Topics"
2. Format as **Heading 1**:
 - **Home** tab → **Styles** → **Heading 1**
 - OR Font: Arial Bold, 16 pt, Dark Blue
3. Press Enter

Create Multi-Level List:

1. Click **Home** tab → **Multilevel List** button
2. Select a numbered list style (e.g., 1, 1.1, 1.1.1)
3. Type the following structure:
 1. First Semester
 - 1.1. FEG-02: English
 - 1.1.1. Grammar and Composition
 - 1.1.2. Reading Comprehension
 - 1.1.3. Business Communication
 - 1.2. ECO-01: Business Organization
 - 1.2.1. Forms of Business Organization
 - 1.2.2. Capital Markets
 - 1.2.3. Banking System
 - 1.3. BCS-011: Computer Basics and PC Software
 - 1.3.1. Computer Fundamentals
 - 1.3.2. Operating Systems
 - 1.3.3. Office Applications
 - 1.4. BCS-012: Basic Mathematics
 - 1.4.1. Matrices and Determinants
 - 1.4.2. Calculus
 - 1.4.3. Vectors
 - 1.5. BCSL-013: Computer Basics Lab
 - 1.5.1. Linux Commands
 - 1.5.2. MS Office Practical
 - 1.5.3. Email Management
 2. Second Semester
 - 2.1. MCS-011: Problem Solving and Programming
 - 2.1.1. Algorithm Design
 - 2.1.2. C Programming Basics
 - 2.1.3. Data Structures
 - 2.2. MCS-012: Computer Organization
 - 2.2.1. Digital Logic
 - 2.2.2. Computer Architecture
 - 2.2.3. Memory Organization
 - 2.3. MCS-013: Discrete Mathematics
 - 2.3.1. Set Theory
 - 2.3.2. Graph Theory
 - 2.3.3. Combinatorics
 - 2.4. BCS-021: Database Management Systems
 - 2.4.1. Relational Database Design
 - 2.4.2. SQL Queries
 - 2.4.3. Normalization
 - 2.5. BCSL-022: Programming Lab
 - 2.5.1. C Programming
 - 2.5.2. Data Structure Implementation
 - 2.5.3. Problem Solving
 3. Third Semester
 - 3.1. MCS-021: Data Structures
 - 3.1.1. Arrays and Linked Lists
 - 3.1.2. Stacks and Queues
 - 3.1.3. Trees and Graphs
 - 3.2. MCS-022: Operating Systems
 - 3.2.1. Process Management
 - 3.2.2. Memory Management
 - 3.2.3. File Systems
 - 3.3. MCS-023: Web Technologies
 - 3.3.1. HTML and CSS
 - 3.3.2. JavaScript
 - 3.3.3. PHP and MySQL
 - 3.4. BCS-031: Programming in C++
 - 3.4.1. Object-Oriented Concepts
 - 3.4.2. Classes and Objects
 - 3.4.3. Inheritance and Polymorphism

****How to Create Multi-Level List:****

1. Type main heading (e.g., "1. First Semester")
2. Press ****Enter****
3. Press ****Tab**** to create sub-level (1.1)
4. Type sub-item
5. Press ****Enter**** and ****Tab**** again for next sub-level (1.1.1)
6. Press ****Enter**** without Tab to stay at same level
7. Press ****Shift+Tab**** to move back to previous level

****Step 4: Create Table****

1. Press Enter twice after the list
2. Type heading: "Course Details"
3. Format as ****Heading 1****
4. Press Enter

****Insert Table:****

1. ****Insert**** tab → ****Table****
2. Select ****4 columns × 11 rows**** (or use Insert Table dialog)
3. OR Type dimensions: 4×11 and click OK

****Table Structure:****

Course Code	Course Name	Credits	Semester
FEG-02	English	4	1
ECO-01	Business Organization	3	1
BCS-011	Computer Basics and PC Software	3	1
BCS-012	Basic Mathematics	3	1
BCSL-013	Computer Basics Lab	2	1
MCS-011	Problem Solving and Programming	4	2
MCS-012	Computer Organization	3	2
MCS-013	Discrete Mathematics	2	2
BCS-021	Database Management Systems	3	2
BCSL-022	Programming Lab	2	2

****Format Table:****

1. ****Select header row**** (first row)
2. ****Table Design**** tab → ****Shading**** → Choose dark blue
3. ****Font color**** → White
4. ****Bold**** the header text
5. Select entire table
6. ****Table Design**** tab → Choose a table style (e.g., "Grid Table 4 - Accent 1")
7. Adjust column widths by dragging borders
8. Center-align Credits and Semester columns:
 - Select these columns
 - ****Layout**** tab → ****Alignment**** → ****Align Center****

****Alternative Formatting:****

1. Right-click table → ****Table Properties****
2. ****Table**** tab → ****Alignment****: Center
3. ****Row**** tab → Specify row height if needed
4. ****Cell**** tab → Vertical alignment: Center

****Step 5: Add Objectives Paragraph****

1. Press Enter twice after table

2. Type heading: "Objectives of BCA Programme"
3. Format as **Heading 1**
4. Press Enter

Type the paragraph:

The Bachelor of Computer Applications (BCA) programme is designed to provide comprehensive knowledge of computer applications and information technology. The primary objectives of the BCA programme are: to equip students with strong fundamentals in computer science and applications; to develop problem-solving and analytical skills through programming and mathematical training; to provide hands-on experience with current technologies and tools used in the IT industry; to prepare students for careers as software developers, system analysts, database administrators, and IT professionals; to create a foundation for higher studies in computer science such as MCA, M.Sc. in Computer Science, or MBA in IT; to foster innovation and creativity in applying technology to solve real-world problems; to develop communication and teamwork skills essential for professional success; and to instill ethical values and professional responsibility in the use of technology. The programme combines theoretical knowledge with practical training through laboratory courses, projects, and internships, ensuring that graduates are industry-ready and capable of contributing effectively to the rapidly evolving field of information technology.

Format Paragraph:

1. **Select paragraph text**
2. **Alignment:** Justify
 - **Home** tab → **Paragraph** section → **Justify** button
3. **Line spacing:** 1.15 or 1.5
 - **Home** tab → **Line and Paragraph Spacing** → **1.5**
4. **Font:** Calibri or Arial, 11 pt
5. Add spacing after paragraph: 6 pt

Alternative - Bulleted Objectives:

The Bachelor of Computer Applications (BCA) programme aims to:

- Provide comprehensive knowledge of computer applications and information technology
- Develop strong programming and problem-solving skills
- Offer hands-on experience with industry-standard tools and technologies
- Prepare students for careers in software development and IT management
- Build a foundation for advanced studies in computer science
- Foster innovation in applying technology to real-world challenges
- Develop communication and teamwork capabilities
- Instill ethical values in technology usage
- Ensure graduates are industry-ready and competitive in the job market
- Create opportunities for research and entrepreneurship in IT

Step 6: Final Document Formatting

1. **Add page numbers:**
 - **Insert** tab → **Page Number** → **Bottom of Page** → Choose style
2. **Add header:**

- **Insert** tab → **Header** → Choose style
- Type: "BCA Programme Information"

3. **Add footer:**

- Already has page number
- Can add date or institution name

4. **Adjust margins** (if needed):

- **Layout** tab → **Margins** → **Normal** (1" all sides)

5. **Apply document theme:**

- **Design** tab → Choose theme (e.g., "Office" or "Facet")

6. **Final review:**

- Check spelling: Press F7
- Check formatting consistency
- Ensure all elements are properly aligned

Part (b): Flyer on "Advantages of Online Banking" (10 Marks)

Design Requirements:

- Different styles, sizes, fonts, colors, and effects
- Attractive and professional appearance
- Clear information hierarchy

Step-by-Step Instructions

Step 1: Document Setup

1. **Create new Word document**
2. Save as "Online_Banking_Flyer.docx"
3. **Set page orientation:**
 - **Layout** tab → **Orientation** → **Landscape** (or Portrait)
4. **Set margins:**
 - **Layout** tab → **Margins** → **Narrow** (0.5" all sides)
5. **Optional - Set page size:**
 - **Layout** tab → **Size** → **A4** or **Letter**

Step 2: Create Flyer Layout

Method 1: Using Built-in Templates

1. **File** → **New**
2. Search for "Flyer"
3. Select appropriate template
4. Customize content

Method 2: Design from Scratch (Recommended for learning)

Add Title Section:

1. **Insert** tab → **Text Box** → **Draw Text Box**
2. Draw large text box at top (spanning width)
3. Type: "ADVANTAGES OF ONLINE BANKING"

4. Format title:
 - Font: **Impact** or **Arial Black**
 - Size: **36-44 pt**
 - Color: **Dark Blue** or **Green** (banking colors)
 - **Bold**
 - Alignment: **Center**
 - Text Effect:
 - **Home** tab → **Text Effects** → **Shadow** or **Reflection**

Step 3: Add Tagline/Subtitle

1. Insert text box below title
2. Type: "Banking Made Easy, Secure, and Convenient - Anytime, Anywhere!"
3. Format:
 - Font: **Calibri** or **Georgia**
 - Size: **16-18 pt**
 - Color: **Dark Gray**
 - Style: **Italic**
 - Alignment: **Center**

Step 4: Add Main Content - Advantages

Create columnar layout:

Advantage 1: 24/7 Accessibility

1. Insert text box (left side)
2. Add icon/image:
 - **Insert** → **Icons** → Search "clock" or "24 hours"
 - OR insert image of 24/7 symbol
3. Type heading: "24/7 Accessibility"
 - Font: **Arial Bold**, 16 pt, Dark Blue
4. Type description:

**Access your account anytime, anywhere. No need to visit branches during banking hours.
Manage finances at your convenience.**

- Font: Calibri, 11 pt, Black

Advantage 2: Quick Transactions

1. Insert text box (right of first)
2. Icon: Money transfer or lightning bolt
3. Heading: "Quick Transactions"
4. Description:

Transfer money instantly between accounts. Pay bills with a click. Real-time transaction updates.

Advantage 3: Lower Costs

1. Insert text box
2. Icon: Piggy bank or coins
3. Heading: "Lower Costs"
4. Description:

Save on paper statements and checks. Reduced transaction fees. No travel costs to branches.

****Advantage 4: Enhanced Security****

1. Insert text box
2. Icon: Shield or lock
3. Heading: "Enhanced Security"
4. Description:

Multi-factor authentication. Encrypted transactions. Real-time fraud alerts and monitoring.

****Advantage 5: Easy Bill Payments****

1. Insert text box
2. Icon: Receipt or document
3. Heading: "Easy Bill Payments"
4. Description:

Schedule automatic payments. Set up recurring transfers. Never miss a payment deadline.

****Advantage 6: Detailed Records****

1. Insert text box
2. Icon: Document or chart
3. Heading: "Detailed Records"
4. Description:

Download statements instantly. Track spending patterns. Access transaction history anytime.

****Step 5: Add Visual Elements****

****Background:****

1. ****Design**** tab → ****Page Color**** → Choose light color (Light Blue or Light Green)
2. OR ****Page Color**** → ****Fill Effects**** → ****Gradient**** (Two colors)

****Shapes for Design:****

1. ****Insert**** tab → ****Shapes****
2. Add rectangles or rounded rectangles behind content boxes
3. Format shapes:
 - ****Shape Fill**** → Complementary color
 - ****Shape Outline**** → No outline or thin border
 - ****Shape Effects**** → Shadow or 3D effect

****Borders:****

1. ****Design**** tab → ****Page Borders****
2. Select ****Box**** border style
3. Choose color and width: 3 pt, Dark Blue
4. Click OK

****Step 6: Add Images/Graphics****

****Main image:****

1. ****Insert**** → ****Pictures**** or ****Online Pictures****
2. Search for:
 - "Online banking concept"

- "Mobile banking"
- "Digital payment"
- 3. Insert and resize (place in center or side)

****Icons for each advantage:****

1. ****Insert**** → ****Icons****
2. Search and insert relevant icons
3. Resize to 1-1.5 inches
4. Change color:
 - Select icon → ****Graphics Format**** → ****Graphics Fill**** → Choose color

****Step 7: Add Call-to-Action Section****

1. Insert text box at bottom
2. Type: "Get Started with Online Banking Today!"
3. Format:
 - Font: Arial Black, 20 pt
 - Color: White
 - Background: Dark Blue rectangle shape behind text
 - Alignment: Center

****Add contact information:****

Visit: www.yourbank.com | Call: 1800-XXX-XXXX | Email: support@yourbank.com

- Font: Calibri, 10 pt, Dark Gray

****Step 8: Apply Advanced Formatting****

****Text Effects:****

1. Select headings
2. ****Home**** → ****Text Effects and Typography****
3. Apply:
 - Gradient Fill
 - Shadow
 - Reflection
 - Glow

****Shape Effects:****

1. Select shapes/text boxes
2. ****Shape Format**** tab → ****Shape Effects****
3. Apply:
 - Shadow (Outer: Offset Right)
 - 3D Rotation
 - Soft Edges

****Color Scheme:****

Use consistent colors throughout:

- ****Primary:**** Dark Blue (#003366)
- ****Secondary:**** Green (#008000)
- ****Accent:**** Orange (#FF8C00)
- ****Background:**** Light Blue (#E6F3FF)
- ****Text:**** Black (#000000) and Dark Gray (#4D4D4D)

****Step 9: Final Touches****

****Add decorative elements:****

1. ****Insert**** → ****Shapes**** → Stars or banners
2. Place strategically for visual appeal
3. Add text inside shapes (e.g., "Save Time!", "Stay Secure!")

****Add footer/disclaimer:****

***Terms and conditions apply. Please visit branch or website for complete details.**

- Font: Calibri, 8 pt, Gray

****Group elements** (optional):**

1. Select multiple objects (hold Ctrl while clicking)
2. Right-click → ****Group**** → ****Group****
3. This allows moving elements together

****Step 10: Review and Export****

1. ****Review document:****

- Check all text is readable
- Verify color contrast
- Ensure proper alignment
- Check for spelling errors (F7)

2. ****Save as PDF:****

- ****File**** → ****Save As****
- Choose ****PDF**** format
- Click ****Save****

Complete Sequence Summary for Q3

****Part (a) - BCA Document:****

1. Create new document and save
2. Add title and format
3. Create multi-level list (courses and topics)
4. Insert and format table (course details)
5. Write and format objectives paragraph
6. Add page numbers and header/footer
7. Apply document theme
8. Save and export as PDF

****Part (b) - Flyer:****

1. Create new document with landscape orientation
2. Set narrow margins
3. Add title with large, bold, colored font
4. Add subtitle with effects
5. Create 6 advantage sections with icons
6. Add main banking image
7. Insert decorative shapes and borders
8. Add call-to-action section
9. Apply consistent color scheme and effects
10. Review and export as PDF

Q4: Excel Worksheet - Sales Data (16 Marks)

Step-by-Step Instructions

Step 1: Open Excel and Setup

1. Open **Microsoft Excel** (or LibreOffice Calc)
2. **File** → **New** → **Blank Workbook**
3. Save as "Sales_Data_Company.xlsx"

Step 2: Enter Headers

Click on cell A1 and type the following headers across Row 1:

Cell	Header Text
A1	Salesperson Name
B1	Annual Sales in Numbers (N)
C1	Price of each Item (P)
D1	Gross Sales in INR (G)
E1	Discount (10% of G) (D)
F1	Net Sales in INR (NS)
G1	GST (10% of NS)
H1	Amount Receivable with GST (R)

Format Headers:

1. Select cells A1:H1
2. **Home** tab → **Font** section:
 - Font: **Arial Bold** or **Calibri Bold**
 - Size: **11 pt**
 - **Bold** (Ctrl+B)
3. **Alignment**:
 - **Wrap Text** (to fit long headers)
 - **Center** alignment
 - **Middle Align** (vertical center)
4. **Fill Color**:
 - **Home** tab → **Fill Color** → Dark Blue
 - **Font Color** → White
5. Adjust row height:
 - Select Row 1
 - Right-click → **Row Height** → Set to 40 or auto-fit

Step 3: Enter Salesperson Data

Enter the following data in columns A, B, and C:

Row	A (Name)	B (Numbers N)	C (Price P)
2	Aana	50000	50
3	Mohan	65000	45
4	Shyama	60000	36
5	Hasan	50000	49
6	Deep	45000	54
7	Roger	30000	65

Format Data Columns:

1. Select cells A2:A7 (Names)
 - Alignment: Left
 - Font: Calibri, 11 pt
2. Select cells B2:C7 (Numbers)
 - Alignment: Right
 - Number format: Number with comma separator
 - Right-click → **Format Cells** → **Number** → **1000 separator**

Step 4: Create Formulas

Column D - Gross Sales ($G = N * P$)

1. Click on cell **D2**
2. Type formula: `=B2*C2`
3. Press **Enter**
4. Result should be: 2,500,000 (for Aana)

Copy formula down:

- Method 1: Click D2, drag fill handle (small square at bottom-right) down to D7
- Method 2: Click D2, press Ctrl+C, select D3:D7, press Ctrl+V
- Method 3: Click D2, double-click fill handle (auto-fills)

Verify results:

- D2: 2,500,000 (50000 × 50)
- D3: 2,925,000 (65000 × 45)
- D4: 2,160,000 (60000 × 36)
- D5: 2,450,000 (50000 × 49)
- D6: 2,430,000 (45000 × 54)
- D7: 1,950,000 (30000 × 65)

Column E - Discount ($D = 10\%$ of G)

1. Click on cell **E2**
2. Type formula: `=D2*0.1`
 - OR: `=D2*10/100`
 - OR: `=D2*10%`
3. Press **Enter**
4. Result: 250,000

Copy formula down to E3:E7

Verify results:

- E2: 250,000
- E3: 292,500
- E4: 216,000
- E5: 245,000
- E6: 243,000
- E7: 195,000

Column F - Net Sales ($NS = G - D$)

1. Click on cell **F2**
2. Type formula: `=D2-E2`

3. Press ****Enter****
4. Result: 2,250,000

****Copy formula down**** to F3:F7

****Verify results:****

- F2: 2,250,000
- F3: 2,632,500
- F4: 1,944,000
- F5: 2,205,000
- F6: 2,187,000
- F7: 1,755,000

****Column G - GST (GST = 10% of NS)****

1. Click on cell ****G2****
2. Type formula: `=F2*0.1``
3. Press ****Enter****
4. Result: 225,000

****Copy formula down**** to G3:G7

****Verify results:****

- G2: 225,000
- G3: 263,250
- G4: 194,400
- G5: 220,500
- G6: 218,700
- G7: 175,500

****Column H - Amount Receivable (R = NS + GST)****

1. Click on cell ****H2****
2. Type formula: `=F2+G2``
3. Press ****Enter****
4. Result: 2,475,000

****Copy formula down**** to H3:H7

****Verify results:****

- H2: 2,475,000
- H3: 2,895,750
- H4: 2,138,400
- H5: 2,425,500
- H6: 2,405,700
- H7: 1,930,500

****Step 5: Format Currency Columns****

1. Select cells ****D2:H7**** (all currency values)
2. Right-click → ****Format Cells****
3. ****Number**** tab → Select ****Currency****
4. Settings:
 - ****Symbol****: ₹ (Indian Rupee)
 - ****Decimal places****: 2

- **Negative numbers**: Red with minus sign
5. Click **OK**

Alternative - Quick Format:

1. Select D2:H7
2. **Home** tab → **Number** section → Click dropdown
3. Select **Currency** or **Accounting**

Step 6: Add Total Row

1. Click on cell **A8**
2. Type: **TOTAL**
3. Format: **Bold**

Add SUM formulas:

Cell B8 (Total Annual Sales Numbers):

=SUM(B2:B7)

Result: 300,000

Cell C8 - Leave blank (price doesn't sum meaningfully)

Cell D8 (Total Gross Sales):

=SUM(D2:D7)

Result: ₹14,415,000.00

Cell E8 (Total Discount):

=SUM(E2:E7)

Result: ₹1,441,500.00

Cell F8 (Total Net Sales):

=SUM(F2:F7)

Result: ₹12,973,500.00

Cell G8 (Total GST):

=SUM(G2:G7)

Result: ₹1,297,350.00

Cell H8 (Total Amount Receivable):

=SUM(H2:H7)

Result: ₹14,270,850.00

Format Total Row:

1. Select A8:H8
2. **Ctrl+B**
3. **Bottom Border** (thick line):
 - **Home** → **Borders** → **Top and Double Bottom Border**

Step 7: Apply Conditional Formatting (Optional Enhancement)

Highlight Top Performers:

1. Select cells H2:H7 (Amount Receivable)
2. **Home** tab → **Conditional Formatting** → **Top/Bottom Rules** → **Top 10 Items**
3. Change to "Top 3"
4. Choose formatting: Light Green Fill
5. Click OK

Color Scale for Sales:

1. Select B2:B7
2. **Conditional Formatting** → **Color Scales**
3. Choose: Red-Yellow-Green scale

Step 8: Add Chart (Optional Enhancement)

1. Select data range: A1:A7 and H1:H7 (Names and Amount Receivable)
2. **Insert** tab → **Charts** → **Column Chart** or **Bar Chart**
3. Chart appears on worksheet
4. **Chart Title**: "Sales Performance by Salesperson"
5. Format chart:
 - **Chart Design** tab → Choose style
 - Add data labels: **Chart Elements** (+) → Check **Data Labels**

Step 9: Adjust Column Widths

Auto-fit columns:

1. Select all columns (click column header A, hold Shift, click H)
2. **Home** tab → **Format** → **AutoFit Column Width**

OR manually adjust:

- Double-click border between column headers to auto-fit
- Drag column borders to desired width

Recommended widths:

- Column A: 15-20 characters
- Columns B-H: 12-15 characters each

Step 10: Add Borders

1. Select entire data range: **A1:H8**
2. **Home** tab → **Borders** dropdown → **All Borders**

Enhance header border:

1. Select A1:H1
2. **Borders** → **Thick Box Border**

Step 11: Final Formatting

Freeze header row:

1. Click on cell A2
2. **View** tab → **Freeze Panes** → **Freeze Top Row**
3. Now header stays visible when scrolling

Center worksheet on page:

1. **Page Layout** tab → **Page Setup** dialog launcher (small arrow)
2. **Margins** tab → **Center on page**: Check **Horizontally**
3. Click OK

Add header/footer:

1. **Insert** tab → **Header & Footer**
2. OR **Page Layout** view → Click header area
3. Type: "Company Sales Data Report - 2025"

Page orientation:

1. **Page Layout** tab → **Orientation** → **Landscape** (recommended)

Step 12: Print Setup

1. **File** → **Print** (or Ctrl+P)
2. **Settings**:
 - Orientation: Landscape
 - Scaling: Fit Sheet on One Page
 - Margins: Normal
3. **Print Preview** - verify appearance
4. Click **Print** or **Save as PDF**

Formula Summary Table

Column	Formula	Example (Row 2)	Purpose
D (Gross Sales)	=B*C	=B2*C2	Sales quantity × Price
E (Discount)	=D*0.1	=D2*0.1	10% of Gross Sales
F (Net Sales)	=D-E	=D2-E2	Gross minus Discount
G (GST)	=F*0.1	=F2*0.1	10% of Net Sales
H (Receivable)	=F+G	=F2+G2	Net Sales + GST
Row 8 (Totals)	=SUM(range)	=SUM(D2:D7)	Sum of column values

Expected Final Worksheet

Salesperson	Annual Sales (P)	Price Sales (G)	Gross (10% G)	Discount Sales NS	Net (10% NS)	GST Receivable(R)	Amount	Name
50,000	50	₹25,00,000	₹2,50,000	₹22,50,000	₹2,25,000	₹24,75,000		Aana Mohan

65,000	45	₹29,25,000	₹2,92,500	₹26,32,500	₹2,63,250	₹28,95,750	Shyama
60,000	36	₹21,60,000	₹2,16,000	₹19,44,000	₹1,94,400	₹21,38,400	Hasan
50,000	49	₹24,50,000	₹2,45,000	₹22,05,000	₹2,20,500	₹24,25,500	Deep
45,000	54	₹24,30,000	₹2,43,000	₹21,87,000	₹2,18,700	₹24,05,700	Roger
30,000	65	₹19,50,000	₹1,95,000	₹17,55,000	₹1,75,500	₹19,30,500	
							TOTAL
300,000		₹1,44,15,000	₹14,41,500	₹1,29,73,500	₹12,97,350	₹1,42,70,850	

Q5: MS Outlook / Email Tasks (16 Marks)

Part (a): Weekly Meeting Schedule (4 Marks)

****Task:**** Create weekly meeting for "Birthday Planning" with 2 friends for 2 consecutive weeks

Step-by-Step Instructions (MS Outlook)

****Step 1: Open Calendar****

1. Open ****Microsoft Outlook****
2. Click on ****Calendar**** icon at bottom (or press Ctrl+2)

****Step 2: Create New Meeting****

1. ****Home**** tab → ****New Meeting**** (or press Ctrl+Shift+Q)
2. OR right-click on calendar date/time → ****New Meeting****

****Step 3: Fill Meeting Details****

****To field:****

- Click ****To**** button
- Select or type email addresses of 2 friends:
 - friend1@email.com
 - friend2@email.com
- Click OK

****Subject:****

Birthday Planning - Weekly Meeting

****Location:****

Conference Room A (or) Online - Zoom Link: <https://zoom.us/j/123456>

****Start time:****

- Date: Select appropriate date (e.g., October 10, 2025)
- Time: 6:00 PM (or suitable time)

****End time:****

- Same date
- Time: 7:00 PM (1 hour duration)

****Message body:****

Hi Team,

This is our weekly meeting to discuss and plan the birthday celebration.

Agenda:

1. Venue selection
2. Guest list finalization
3. Decoration ideas
4. Food menu planning
5. Budget allocation
6. Task assignments

Please come prepared with your ideas!

Looking forward to our discussion.

Best regards, [Your Name]

****Step 4: Set Recurrence****

1. Click ****Recurrence**** button in ****Options**** section (ribbon)
2. ****Recurrence pattern**** dialog opens
3. Select:
 - ****Recurrence pattern****: Weekly
 - ****Recur every****: 1 week
 - ****Days****: Check the day (e.g., Friday)
4. ****Range of recurrence****:
 - ****Start****: October 10, 2025
 - Select ****End after****: 2 occurrences
5. Click ****OK****

****Step 5: Set Reminder****

1. ****Reminder****: 15 minutes (or 1 hour before)
2. Check ****Request Responses**** (to track attendance)

****Step 6: Send Meeting Invitation****

1. Click ****Send**** button
2. Meeting invitations sent to both friends
3. Meeting appears in your calendar with recurrence icon

****Verification:****

- Check calendar - should show 2 meetings (Week 1 and Week 2)
- Both dates should be exactly 1 week apart
- Recurrence icon should appear on meeting

****Alternative - Google Calendar:****

1. Open ****Google Calendar**** (calendar.google.com)

2. Click **Create** or click on date/time
3. Click **Add title**: "Birthday Planning - Weekly Meeting"
4. **Add guests**: Enter friend emails
5. Set **date and time**
6. Click **Does not repeat** → Select **Weekly**
7. Set **Ends**: After 2 occurrences
8. Add location and description
9. Click **Save** → **Send** invitations

Part (b): Birthday Invitation Email with Signature (4 Marks)

Task: Email to 5 friends inviting to birthday with custom signature

Step 1: Create Email Signature First

In MS Outlook:

1. Click **File** → **Options**
2. Select **Mail** in left pane
3. Click **Signatures** button
4. Click **New** in Signatures dialog
5. Name it: "Personal_Signature"
6. In edit box, type:

Warm Regards, [Your Name]

✉ Email: yourname@email.com 📞 Phone: +91-9876543210 🌐 Website:
www.yourwebsite.com 🍰 Let's celebrate together!

"Life is a party, dress like it!" - Audrey Hepburn

Format signature:

1. Select your name → Make it **Bold** and larger (14 pt)
2. Select quote → Make it *Italic* and smaller (9 pt)
3. Add line separator: Insert → Horizontal Line

Add image/logo (optional):

1. Click where you want image
2. **Insert** → **Picture**
3. Choose small image (birthday cake icon, your photo)
4. Resize to 100×100 pixels or smaller

Save signature:

1. Click **OK** to save
2. Set as **default signature** for new messages
3. Click **OK** to close Options

Alternative - Create HTML Signature:

```
```html
```

```
<div style="font-family: Calibri, Arial, sans-serif; font-size: 11pt;">
```

```
<p style="margin: 0; padding: 0;">
 <strong style="color: #0066cc; font-size: 14pt;">Warm
Regards,

 <strong style="font-size: 12pt;">[Your Full Name]
</p>
<p style="margin: 5px 0; color: #666;">
 ✉ yourname@email.com | 📞 +91-9876543210

 🌐 www.yourwebsite.com

 🎉 Let's celebrate together!
</p>
<p style="font-style: italic; font-size: 9pt; color: #888;">
 "Life is a party, dress like it!" - Audrey Hepburn
</p>
</div>
```

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## Step 2: Compose Birthday Invitation Email

### 1. Home tab → New Email (Ctrl+N)

#### To:

friend1@email.com; friend2@email.com; friend3@email.com; friend4@email.com;  
friend5@email.com

#### Subject:

🎉 You're Invited to My Birthday Celebration! 🎂

#### Email Body:

Dear Friends,

I hope this email finds you in great spirits!

I am thrilled to invite you to my Birthday Celebration, and I would be honored to have you join me for this special occasion.

📅 Date: Saturday, January 25, 2026  
🕒 Time: 6:00 PM onwards  
📍 Venue: The Grand Ballroom, Hotel Paradise  
123 Main Street, [Your City]  
👔 Dress Code: Smart Casual / Party Wear

#### EVENT HIGHLIGHTS:

- Welcome drinks and appetizers
- Delicious dinner buffet
- Live music and DJ
- Games and entertainment
- Photo booth corner
- Dance floor

It would mean the world to me to celebrate this special day surrounded by wonderful people like you. Your presence will make the celebration complete!

Please RSVP by January 15, 2026, so we can make necessary arrangements.

Feel free to bring along a plus one if you'd like!

Looking forward to celebrating with you! ☐

[Signature will appear here automatically]

### **Format email:**

1. Select subject line emojis (if supported)
  2. Make event details **Bold**
  3. Format date/time/venue in a different color (Blue or Dark Red)
  4. Add bullet points for highlights
  5. Use appropriate font size (11-12 pt)
- 

### **Step 3: Add Signature**

1. If signature doesn't appear automatically:
    - **Insert** tab → **Signature** → Select "Personal\_Signature"
  2. Signature appears at bottom of email
- 

### **Step 4: Enhance Email (Optional)**

#### **Add image:**

1. **Insert** tab → **Pictures**
2. Choose birthday-themed image
3. Resize and position appropriately

#### **Add invitation card attachment:**

1. **Attach File** → Browse
2. Attach designed invitation card (PDF or image)

#### **Set importance:**

1. **Message** tab → **High Importance** flag

#### **Request Read Receipt:**

1. **Options** tab → Check **Request a Read Receipt**
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### **Step 5: Send Email**

1. Review email for errors
2. Verify all 5 email addresses in To field
3. Ensure signature is present
4. Click **Send**

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### Screenshot Requirements:

- Signature creation dialog
- Composed email showing all fields
- Email with signature visible
- Sent items showing the email

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### Part (c): Auto-Reply Message for January 1, 2026 (4 Marks)

**Task:** Create automatic reply for all emails received on January 1, 2026

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### Step-by-Step Instructions

#### Method 1: Out of Office Auto-Reply

1. **File** → **Info**
2. Click **Automatic Replies (Out of Office)**
3. Select **Send automatic replies**
4. **Time range:**
  - **Start time:** January 1, 2026, 12:00 AM
  - **End time:** January 1, 2026, 11:59 PM
5. **Inside My Organization** tab:

Subject: Happy New Year 2026! 🎉

Dear Sender,

Thank you for your email.

I am currently unavailable as I am celebrating New Year's Day with family and friends. I will have limited access to email throughout the day.

I will respond to your message on January 2, 2026, when I return to regular work schedule.

For urgent matters, please contact:

✉️ urgent@company.com

📞 +91-9999888877

Wishing you a wonderful start to 2026!

Best regards,



[Your Name]

6. **Outside My Organization** tab (optional):
    - Check **Send automatic replies to people outside my organization**
    - Copy same message or customize
  7. Click **OK**
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## Method 2: Rules-Based Auto-Reply (More Flexible)

1. **Home** tab → **Rules** → **Manage Rules & Alerts**
2. Click **New Rule**
3. Select **Apply rule on messages I receive**
4. Click **Next**

**Conditions:** 5. Check **received in a specific date span** 6. Click underlined text in description 7. Set:

- **After:** 1/1/2026 12:00 AM
- **Before:** 1/1/2026 11:59 PM

8. Click **Next**

**Actions:** 9. Check **reply using a specific template** 10. Click underlined text 11. Click **New** to create template

**Create Template Message:** 12. In new message window: - **Subject:** Happy New Year 2026! - **Body:** (Same message as above) 13. **File** → **Save As** 14. **Save as type:** Outlook Template (\*.oft) 15. Name: "NewYear\_AutoReply" 16. Save to default Templates folder 17. Close template message

**Continue Rule Setup:** 18. Select the template you just created 19. Click **Open** → **Next** 20. Add exceptions if needed (e.g., don't reply to certain domains) 21. Click **Next** 22. Name rule: "New Year 2026 Auto-Reply" 23. Check **Turn on this rule** 24. Click **Finish**

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## Method 3: Quick Steps (for recurring use)

1. **Home** tab → **Quick Steps** → **New Quick Step**
  2. Name: "New Year Auto-Reply"
  3. **Choose an action:** Reply
  4. Click **Add Action** → **Reply**
  5. Configure message template
  6. Assign keyboard shortcut
  7. Click **Finish**
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### Alternative - Gmail Auto-Responder:

1. **Settings** (gear icon) → **See all settings**
  2. Scroll to **Vacation responder**
  3. Select **Vacation responder on**
  4. **First day:** January 1, 2026
  5. **Last day:** January 1, 2026
  6. **Subject:** Happy New Year 2026!
  7. **Message:** (Same content)
  8. Optional: **Send a response only to people in my Contacts**
  9. Click **Save Changes**
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### Verification:

- Send test email to yourself after setting up
  - Check if auto-reply is received
  - Verify date range is correct
  - Test before actual date
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### Part (d): Block/Auto-Delete Emails from Specific Sender (4 Marks)

**Task:** Automatically delete emails from a person sending multiple unwanted emails

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### Step-by-Step Instructions

#### Method 1: Block Sender (Recommended)

1. **Open any email** from the unwanted sender
2. **Home** tab → **Junk** → **Block Sender**
3. Confirmation dialog appears:
  - "The sender will be added to Blocked Senders list"
  - "All current and future messages from this sender will be sent to Junk Email"
4. Click **OK**

#### Alternative access:

- Right-click on email → **Junk** → **Block Sender**

#### Manage Blocked Senders:

1. **Home** tab → **Junk** → **Junk E-mail Options**
2. **Blocked Senders** tab
3. View list of blocked addresses
4. Add, Edit, or Remove senders

5. Click **OK**

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## Method 2: Create Rule to Auto-Delete

1. **Home** tab → **Rules** → **Manage Rules & Alerts**
2. Click **New Rule**
3. Select **Apply rule on messages I receive**
4. Click **Next**

**Step 1: Conditions** 5. Check **from people or public group** 6. In description, click underlined **people or public group** 7. Type or select unwanted sender's email: unwanted@email.com 8. Click **From** → **OK** 9. Click **Next**

**Step 2: Actions** 10. Check **permanently delete it** - OR **delete it** (moves to Deleted Items) - OR **move it to the specified folder** → Select **Deleted Items** 11. Click **Next**

**Step 3: Exceptions (optional)** 12. Add exceptions if needed (e.g., if subject contains specific words, don't delete) 13. Click **Next**

**Step 4: Finish Rule** 14. Name rule: "Auto-Delete from [Sender Name]" 15. Check **Turn on this rule** 16. Optional: Check **Run this rule now on messages already in the current folder** 17. Click **Finish**

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## Method 3: Advanced Rule (Multiple Conditions)

**For multiple senders or domains:**

1. Create rule as above
2. In conditions, add multiple addresses:
  - Click **people or public group**
  - Add: spam1@email.com
  - Click **From** → **OK**
  - Repeat to add more addresses

**For entire domain:**

1. In conditions, check **with specific words in the sender's address**
2. Click underlined link
3. Type: @spammer.com
4. Click **Add** → **OK**
5. All emails from @spammer.com domain will be affected

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## Method 4: Junk Email Filter Settings

### Configure aggressive filtering:

1. **Home** tab → **Junk** → **Junk E-mail Options**
2. **Options** tab
3. Select protection level:
  - **Low** (moves obvious spam)
  - **High** (catches more but may have false positives)
  - **Safe Lists Only** (only accepts from Safe Senders list)
4. Check **Permanently delete suspected junk e-mail**
5. Click **OK**

**Warning:** This deletes without moving to Junk folder - cannot recover

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### Method 5: Sweep Function (Outlook.com/Web)

1. Right-click on email from unwanted sender
  2. Select **Sweep**
  3. Choose option:
    - Delete all from sender
    - Delete future messages from sender
    - Keep latest and delete rest
  4. Click **OK**
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### Best Practice Approach:

#### Step 1: Block Sender

- Moves current and future emails to Junk Email folder

#### Step 2: Create Delete Rule

- Automatically permanently deletes emails from blocked sender

### Combined Configuration:

1. Block sender (moves to Junk)
2. Create rule:  
Condition: From unwanted@email.com  
Action: Permanently delete it

This ensures immediate deletion without cluttering Junk folder.

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### Verification:

1. **Test the rule:**

- Send test email from blocked address (use another account)
  - Check if it's automatically deleted
  - Verify in Deleted Items or not appearing at all
  - 2. **Monitor rules:**
    - **Rules** → **Manage Rules & Alerts**
    - Verify rule is enabled (checkbox checked)
    - Check rule description
  - 3. **Review periodically:**
    - Ensure no legitimate emails blocked accidentally
    - Update blocked senders list as needed
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## **Screenshot Requirements for Q5:**

### **Part (a) - Meeting Schedule:**

- Meeting invitation dialog showing recurrence settings
- Calendar view showing 2 recurring meetings
- Sent meeting confirmation

### **Part (b) - Email Invitation:**

- Signature creation dialog
- Composed email with all fields filled
- Email showing signature at bottom
- List of 5 recipients in To field

### **Part (c) - Auto-Reply:**

- Out of Office settings dialog showing date range
- Auto-reply message preview
- Confirmation that auto-reply is enabled

### **Part (d) - Block Sender:**

- Junk Email Options showing blocked senders
  - Rule creation dialog showing delete action
  - Rules list showing the new rule enabled
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